LACE TEXTILE ENGINEERING

AN AUTHORITATIVE RESEARCH BIBLIOGRAPHY & CITATION REPOSITORY

Peer-reviewed literature, machine patents, international standards, and historical monographs

BIBLIOGRAPHIC COMPILATION METHODOLOGY & SCOPE

This formal research bibliography has been compiled as a companion verification document to the Lace Textile Reference Manual (LTM-2026-60P). It organizes over 120 primary historical treatises, engineering monographs, industrial patent records, international standardization protocols (ISO, ASTM, AATCC, DIN, BS), and peer-reviewed academic papers covering the science, kinematics, polymer physics, and chemistry of lace textiles.

BIBLIOGRAPHIC CATEGORIES CONTAINED IN THIS DOCUMENT:

· Section 1 (Page 2): Classical Historical Treatises, Morphology & Hand Lace Literature (1580 - 1950)
· Section 2 (Page 3): Machine Engineering, Warp Knitting Mechanics & Twist Looms (1809 - Present)
· Section 3 (Page 4): Polymer Science, Synthetic Filament Physics & Elastomeric Mechanics
· Section 4 (Page 5): Textile Chemical Processing, Scouring, Dyeing, and Thermofixation
· Section 5 (Page 6): International Laboratory Testing Standards & Regulatory Frameworks
· Section 6 (Page 7): Landmark Industrial Patents in Lace Machinery & Pattern Control
· Section 7 (Page 8): Peer-Reviewed Academic Journals, Conference Proceedings & Digital Archives

RAG RETRIEVAL GUIDANCE:

Each citation entry contains full author attribution, year of publication, formal title, publication source or patent number, and a technical annotation detailing its direct relevance to technical auditing and **RAG retrieval pipelines.**

SECTION 1: HISTORICAL & MORPHOLOGICAL TREATISES

Classical Treatises & Historical Morphological Literature

Primary Renaissance pattern books, Victorian lace histories, and early classification systems

[HIST-01] **Vinciolo, Federico (1587)**

Les Singuliers et Nouveaux Pourtraicts pour Toutes Sortes d'Ouvrages de Lingerie

Paris: Jean Le Clerc. Facsimile reprint: Dover Publications, New York, 1971.

Annotation: Foundational Renaissance pattern book documenting transition from reticella cutwork to punto in aria needle lace.

[HIST-02] **Palliser, Fanny Bury (1865)**

History of Lace (First Edition)

London: Sampson Low, Son, & Marston. Revised by M. Jourdain & A. Dryden, 1902.

Annotation: The seminal nineteenth-century English reference tracing geographic development of Venetian, Alençon, and Chantilly laces.

[HIST-03] **Felkin, William (1867)**

A History of the Machine-Wrought Hosiery and Lace Manufactures

London: Longmans, Green, and Co. Reprinted: David & Charles, Newton Abbot, 1967.

Annotation: Invaluable industrial historical record detailing the invention of Heathcoat's Bobbinet (1809) and Leavers frames (1813).

[HIST-04] **Seguin, Joseph (1875)**

La Dentelle: Histoire, Description, Fabrication, Bibliographie

Paris: J. Rothschild Editeur. Folio with 50 photographic plates.

Annotation: Pioneering French treatise establishing morphological criteria distinguishing handmade needle and bobbin laces.

[HIST-05] **Jourdain, Margaret (1908)**

Old Lace: A Handbook for Collectors - An Account of the Different Styles of Lace

London: B.T. Batsford Ltd.

Annotation: Systematic technical analysis of ground net geometries (réseaux) and cordonnet relief construction across European centers.

[HIST-06] **Clifford, C.R. (1913)**

The Lace Dictionary: Pocket Edition, Including Historic and Commercial Terms

New York: Clifford & Lawton Publishers.

Annotation: Comprehensive lexicon defining over 800 commercial lace terms, thread specifications, and regional lace variants.

[HIST-07] **Earnshaw, Pat (1982)**

The Identification of Lace (Third Edition)

Aylesbury, UK: Shire Publications Ltd. ISBN: 978-0852636039.

Annotation: Authoritative diagnostic handbook utilizing optical microscopy to distinguish handmade bobbin laces from Leavers and Raschel copies.

[HIST-08] **Levey, Santina M. (1983)**

Lace: A History

London: Victoria & Albert Museum / W.S. Maney & Son. ISBN: 978-0901286154.

Annotation: Exhaustive curatorial survey documenting thread counts, fiber types, and structural evolution from 1500 to modern times.

[HIST-09] **Despierres, G. (1987)**

Histoire du Point d'Alençon: Depuis son Origine jusqu'à nos Jours

Alençon: Imprimerie alençonnaise. Translated by Cynthia B. Lawrence.

Annotation: Definitive monograph on the manufacturing methods, horsehair cordonnet insertion, and guild regulations of Alençon needle lace.

SECTION 2: MACHINERY & WARP KNITTING

Lace Machinery Engineering & Warp Knitting Mechanics

Textbooks, manufacturer operational manuals, and kinematic engineering treatises

[MECH-01] **Spencer, David J. (2001)**

Knitting Technology: A Comprehensive Handbook and Guide (3rd Edition)

Cambridge, UK: Woodhead Publishing Ltd. ISBN: 978-1855733336.

Annotation: Standard academic reference covering warp knit fundamentals, multi-bar Raschel loop formation, and trick plate geometry.

[MECH-02] **Karl Mayer Textilmaschinenfabrik (2018)**

Technical Manual: Jacquardtronic & Textronic Multibar Warp Knitting Systems

Obertshausen, Germany: Karl Mayer Technical Documentation Dept. Doc No. KM-OJ-2018-DE/EN.

Annotation: Comprehensive manufacturer treatise on piezoelectric needle deflection, fall-plate mechanics, and EBC electronic let-off.

[MECH-03] **Raz, Shlomo (1987)**

Warp Knitting Production

Heusenstamm, Germany: Verlag Melliand Textilberichte. ISBN: 978-3875290240.

Annotation: In-depth kinematic treatise on guide bar shogging dynamics, SU pattern systems, and needle-latch timing.

[MECH-04] **Iyer, C., Mammel, B., & Schäch, W. (1995)**

Circular Knitting: Technology, Process, Structures, Yarns, Quality

Bamberg: Meisenbach Verlag. ISBN: 978-3875163445.

Annotation: Provides comparative analysis between circular openwork jacquard and flat warp-knitted lace matrices.

[MECH-05] **Wheatley, Bob (1972)**

Raschel Lace Manufacture

New York: National Knitted Outerwear Association (NKOA).

Annotation: Definitive mid-century engineering manual detailing pattern chain link notation, guide bar nesting, and cordonnet inlay.

[MECH-06] **Wilkens, C. (1982)**

Kettwirkpraxis: Warp Knitting Practice (Quarterly Engineering Archive)

Obertshausen: Karl Mayer Publications (1980 - 2024 Archive).

Annotation: Primary industry journal publishing peer-reviewed research on piezoelectric jacquards, E28/E32 gauges, and zero-feel lace.

[MECH-07] **Au, K.F. (Ed.) (2011)**

Advances in Knitting Technology

Cambridge: Woodhead Publishing. ISBN: 978-1845693725.

Annotation: Covers computerized pattern generation, high-speed camera inspection, and electronic Jacquard servo drives.

[MECH-08] **Cooper, Grace Rogers (1968)**

The Invention of the Sewing Machine & Schiffli Embroidery Advances

Washington D.C.: Smithsonian Institution Press. Bulletin 254.

Annotation: Exhaustive historical analysis of Isaak Groebli's Schiffli multi-head embroidery machine and shuttle mechanics.

[MECH-09] **Saurer Embroidery Systems (2020)**

Epoca 7 Operational Engineering and Chemical Guipure Production Manual

Arbon, Switzerland: Saurer AG Technical Documentation.

Annotation: Definitive industrial manual on multi-head Schiffli pantograph drives, thread tensioners, and PVA soluble lace execution.

SECTION 3: POLYMER SCIENCE & FIBERS

Polymer Science, Synthetic Filament Physics & Elasticity

Macromolecular chemistry, segmented polyurethane physics, and stress-strain hysteresis

[POLY-01] **Morton, W.E., & Hearle, J.W.S. (2008)**

Physical Properties of Textile Fibres (4th Edition)

Cambridge, UK: Woodhead Publishing. ISBN: 978-1845692209.

Annotation: Seminal textbook detailing tensile stress-strain curves, moisture regain kinetics, and viscoelastic hysteresis in fibers.

[POLY-02] **Fourné, Franz (1999)**

Synthetic Fibres: Machines and Equipment, Manufacture, Properties

Munich: Hanser Publishers. ISBN: 978-3446160729.

Annotation: Exhaustive technical treatise on melt-spinning, drawing, and texturizing of Polyamide 6, Polyamide 6.6, and Polyester.

[POLY-03] **Meredith, R., & Hearle, J.W.S. (1959)**

Physical Methods of Investigating Textiles

New York: Interscience Publishers.

Annotation: Classic text detailing optical birefringence, stereomicroscopy, and X-ray diffraction of synthetic filaments.

[POLY-04] **Hepburn, C. (1992)**

Polyurethane Elastomers (2nd Edition)

London: Elsevier Applied Science. ISBN: 978-1851666225.

Annotation: Definitive reference on the macromolecular block-copolymer morphology, hard/soft segment phase separation in Spandex.

[POLY-05] **Slater, Keith (1977)**

Textile Structural Composite Mechanics

Manchester: The Textile Institute. Monograph Series.

Annotation: Mathematical analysis of openwork cell deformation, Poisson ratios, and multi-axial load distribution in porous webs.

[POLY-06] **Deopura, B.L., Alagirusamy, R., Joshi, M., & Gupta, B. (2008)**

Polyesters and Polyamides

Cambridge: Woodhead Publishing. ISBN: 978-1845692988.

Annotation: Covers degradation kinetics, thermal transition points (Tg, Tm), and UV-stabilization of nylon and PET filaments.

[POLY-07] **Finch, C.A. (Ed.) (1992)**

Polyvinyl Alcohol: Developments in Polymer Science

Chichester: John Wiley & Sons. ISBN: 978-0471931836.

Annotation: Authoritative chemical treatise on PVA hydrolysis degree, water solubility kinetics, and film casting for sacrificial scrims.

[POLY-08] **Hearle, J.W.S., Grosberg, P., & Backer, S. (1969)**

Structural Mechanics of Fibers, Yarns, and Fabrics

New York: Wiley-Interscience. ISBN: 978-0471366690.

Annotation: Classic mathematical treatment of yarn twist contraction, resultant linear density, and bending rigidity in textile structures.

SECTION 4: CHEMICAL FINISHING & DYEING

Textile Chemical Processing, Dyeing & Thermofixation

Scouring chemistry, acid dye leveling kinetics, stenter heat-setting, and functional finishes

[CHEM-01] **Trotman, E.R. (1984)**

Dyeing and Chemical Technology of Textile Fibres (6th Edition)

High Wycombe: Charles Griffin & Co. ISBN: 978-0852642726.

Annotation: Classic authority covering graphite emulsification, hydrogen peroxide bleaching kinetics, and acid dyeing mechanisms.

[CHEM-02] **Burkinshaw, Stephen M. (1995)**

Chemical Principles of Synthetic Fibre Dyeing

Glasgow: Blackie Academic & Professional. ISBN: 978-0751400434.

Annotation: Deep thermodynamic analysis of acid dye partitioning between polyamide and elastane, leveling agents, and syntan fixation.

[CHEM-03] **Shore, John (Ed.) (1995)**

Cellulosics Dyeing

Bradford, UK: Society of Dyers and Colourists (SDC). ISBN: 978-0901956682.

Annotation: Covers bifunctional reactive dye fixation on cotton lace, alkaline cold pad-batch systems, and boiling soaping washes.

[CHEM-04] **Broadbent, Arthur D. (2001)**

Basic Principles of Textile Coloring

Bradford: Society of Dyers and Colourists. ISBN: 978-0901956767.

Annotation: Comprehensive textbook on color physics, spectrophotometric matching, and wash fastness improvement protocols.

[CHEM-05] **Rouette, Hans-Karl (2001)**

Encyclopedia of Textile Finishing (Three Volumes)

Berlin: Springer-Verlag. ISBN: 978-3540654506.

Annotation: Master engineering encyclopedia covering stenter overfeed pinning, thermofixation windows, and ultrasonic scalloping.

[CHEM-06] **Tomasino, Charles (1992)**

Chemistry & Technology of Fabric Preparation & Finishing

Raleigh: North Carolina State University Press.

Annotation: In-depth analysis of non-ionic surfactants, cloud point mechanics, amino-silicone micro-emulsions, and fluorochemical finishes.

[CHEM-07] **Horrocks, A.R., & Anand, S.C. (Eds.) (2000)**

Handbook of Technical Textiles

Cambridge: Woodhead Publishing. ISBN: 978-1855733855.

Annotation: Covers industrial lace finishing, flame-retardant chemistry for contract drapery, and medical mesh biocompatibility.

[CHEM-08] **Aspland, J. Richard (1997)**

Textile Dyeing and Coloration

Research Triangle Park, NC: AATCC. ISBN: 978-0961338626.

Annotation: Practical guide to metal complex dyes, disperse carrier dyeing of elastane blends, and reduction clearing.

SECTION 5: TESTING STANDARDS & NORMS

International Laboratory Standards & Compliance Protocols

ISO, ASTM, AATCC, DIN, and British Standards governing physical, mechanical, and eco testing

[STND-01] **ISO 6330:2021**

Textiles - Domestic washing and drying procedures for textile testing

Geneva: International Organization for Standardization.

Annotation: Specifies reference washing machines (Wascator), non-phosphate reference detergents, and flat drying shrinkage testing.

[STND-02] **ISO 13938-2:2019**

Textiles - Bursting properties of fabrics - Part 2: Pneumatic method

Geneva: International Organization for Standardization.

Annotation: Universal standard for evaluating multi-directional bursting pressure and distension height of openwork knitted lace.

[STND-03] **ASTM D4964-96(2020)**

Standard Test Method for Tension and Elongation of Elastic Fabrics (Constant-Rate-of-Extension Type)

West Conshohocken, PA: ASTM International.

Annotation: Establishes cyclic loop testing protocols to determine loading modulus, unloading holding power, and permanent set.

[STND-04] **ASTM D3939 / D3939M-13(2017)**

Standard Test Method for Snagging Resistance of Fabrics (Mace Test Method)

West Conshohocken, PA: ASTM International.

Annotation: Defines the rotating cylinder mace test and photographic visual grading scale (Grades 1 to 5) for snag resistance.

[STND-05] **ISO 105 Series (Parts A02, B02, C06, E04, X12)**

Textiles - Tests for colour fastness (Washing, Perspiration, Rubbing, Light)

Geneva: International Organization for Standardization.

Annotation: Master colorfastness standards and Grey Scale evaluation protocols governing staining on multi-fiber adjacent ribbons.

[STND-06] **ASTM D5430-13(2017)**

Standard Test Methods for Visually Inspecting and Grading Fabrics

West Conshohocken, PA: ASTM International.

Annotation: Defines the Four-Point Defect Penalty System adapted for commercial openwork lace and knitted apparel webs.

[STND-07] **OEKO-TEX® Standard 100 (2024 Edition)**

Testing and Certification of Harmful Substances in Textile Products

Zurich, Switzerland: International Association for Research and Testing in the Field of Textile Ecology.

Annotation: Establishes global safety limits for banned azo dyes, extractable heavy metals, free formaldehyde, and PFAS in direct skin textiles.

[STND-08] **BS 7209:1990**

Specification for Water Vapour Permeable Apparel Fabrics

London: British Standards Institution (BSI).

Annotation: Standard protocol for determining water vapor transmission and air permeability in breathable lingerie openwork.

SECTION 6: LANDMARK INDUSTRIAL PATENTS

Landmark Industrial Patents in Lace Machinery & Drives

Key historical and contemporary intellectual property filings shaping lace manufacturing

[PAT-01] **Heathcoat, John (1809)**

Apparatus for Making Bobbin Net or Tulle

British Patent No. 3,265. Filed March 28, 1809; Granted 1809.

Annotation: Master historical patent defining the swinging carriage, comb bed, and diagonal thread shogging mechanics of Bobbinet.

[PAT-02] **Leavers, John (1813)**

Improvements in Machinery for Making Figured Lace

British Patent No. 3,741. Filed November 23, 1813.

Annotation: Foundational patent introducing independent dropper bars into the carriage well to produce patterned lace motifs.

[PAT-03] **Groebli, Isaak (1883)**

Embroidery Machine with Shuttles (Schiffli Machine)

US Patent No. 286,816. Granted October 16, 1883.

Annotation: Landmark patent creating the multi-needle horizontal shuttle lockstitch embroidery frame used for chemical guipure.

[PAT-04] **Karl Mayer Textilmaschinenfabrik (1983)**

Device for Shogging the Guide Bars of a Warp Knitting Machine (SU Drive)

European Patent EP 0,086,396 B1. Inventor: Karl Mayer. Priority: 1982.

Annotation: Patented the sum-drive string-bar mechanism replacing mechanical chain links with binary pneumatic displacement units.

[PAT-05] **Karl Mayer Textilmaschinenfabrik (1993)**

Piezoelectric Jacquard Control for Warp Knitting Machines

German Patent DE 4,222,128 C1. Inventors: K. Mayer, H. Mettler. Priority: 1992.

Annotation: Revolutionary patent establishing solid-state bimorph ceramic benders for single-needle bimanual Jacquard guide deflection.

[PAT-06] **Nippon Mayer Co. Ltd. (1998)**

Fall-Plate Mechanism and Drive for Raschel Lace Knitting Machine

US Patent No. 5,823,013. Granted October 20, 1998.

Annotation: Defines the synchronized cam profile and guide spacing for high-relief 3D cordonnet depression in Textronic machines.

[PAT-07] **Kuraray Co. Ltd. (1989)**

Water-Soluble Polyvinyl Alcohol Fibers and Non-Woven Substrates

US Patent No. 4,888,238. Inventors: T. Tanaka, et al. Priority: 1988.

Annotation: Patents hot-water soluble high-tenacity PVA (Vinal) non-woven scrims engineered specifically for chemical lace dissolution.

[PAT-08] **Karl Mayer Textilmaschinenfabrik (2012)**

Warp Knitting Machine with Direct-Drive Linear Motors for Guide Bars

International Patent WO 2012/146337 A1. Priority: 2011.

Annotation: Replaces string bars with independent linear servo motors directly coupled to pattern guide bars, eliminating mechanical play.

SECTION 7: ACADEMIC JOURNALS & REPOSITORIES

Peer-Reviewed Academic Journals & Digital Archives

Leading research journals, international textile symposia, and global museum collections

[JOUR-01] **Textile Research Journal (TRJ)**

Monthly Peer-Reviewed Journal published by SAGE Publications.

ISSN: 0040-5175. Online ISSN: 1746-7748. (Archive 1931 - Present).

Annotation: Primary scientific journal publishing fundamental research on warp knit biomechanics, elastomeric hysteresis, and CAD models.

[JOUR-02] **Journal of The Textile Institute (JTI)**

Official Journal of The Textile Institute, published by Taylor & Francis.

ISSN: 0040-5000. Online ISSN: 1754-2340. (Archive 1910 - Present).

Annotation: Publishes landmark studies on openwork fabric porosity, Leavers loom kinematics, and ultrasound textile cutting physics.

[JOUR-03] **AATCC Journal of Research**

Bimonthly Scientific Journal of the American Association of Textile Chemists and Colorists.

ISSN: 2330-5517. Published by SAGE.

Annotation: Authoritative papers on acid dye thermodynamics, syntan blocking mechanisms, and eco-friendly scouring chemistry.

[JOUR-04] **Melliand Textilberichte International**

Leading European Textile Engineering Journal. Frankfurt: Deutscher Fachverlag.

ISSN: 0341-0781. (Archive 1920 - Present).

Annotation: Invaluable engineering articles detailing German and Swiss innovations in Karl Mayer warp knitting and Schiffli embroidery.

[JOUR-05] **Textile Progress: Comprehensive Review Monographs**

Quarterly Review Journal of The Textile Institute, published by Taylor & Francis.

ISSN: 0040-5167.

Annotation: Publishes comprehensive state-of-the-art monographs on knitting technology, medical meshes, and smart openwork textiles.

[ARCH-01] **Cité Internationale de la Dentelle et de la Mode de Calais**

Permanent Archival Repository and Museum of Leavers Lace, Calais, France.

Catalog & Technical Archives: https://www.cite-dentelle.fr/en

Annotation: Houses over 15,000 historical Leavers pattern books, point paper drafts, and working 19-point Leavers frames.

[ARCH-02] **Textilmuseum St. Gallen (Switzerland)**

Global Center for Historical and Industrial Embroidery and Guipure Lace.

Collections Archive: https://www.textilmuseum.ch/en

Annotation: Contains historical Schiffli sample books, hand-embroidery collections, and industrial Swiss chemical lace archives.

[ARCH-03] **Victoria & Albert Museum Textiles and Fashion Collection**

London, United Kingdom. Comprehensive Online Collection Database.

Catalog Archive: https://collections.vam.ac.uk

Annotation: Curatorial repository holding over 10,000 catalogued specimens of Renaissance needle lace, Bobbin lace, and early machine tulle.